

CHEMICAL ANALYSIS AND DATA HANDLING

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week ④

④ 1.1. INTRODUCTION TO ANALYTICAL CHEMISTRY

(Analytical chemistry studies and uses instruments and methods used to separate, identify, and quantify matter.) It involves the application of a range of techniques and methodologies to obtain and assess qualitative, quantitative and structural information on the nature of matter.

- **Qualitative analysis** is the identification of elements, species and/or compounds present in a **sample**.
- **Quantitative analysis** is the determination of the absolute or relative amounts of elements, species or compounds present in a **sample**.
- **Structural analysis** is the determination of the spatial arrangement of atoms in an element or molecule or the identification of characteristic groups of atoms (functional groups).
- An element, species or compound that is the subject of analysis is known as an **analyte**.
- The remainder of the material or sample of which the analyte(s) form(s) a part is known as the **matrix**.

The gathering and interpretation of qualitative, quantitative and structural information is essential to many aspects of human endeavor, both terrestrial and extra-terrestrial. The maintenance of, and improvement in, the quality of life throughout the world, and the management of resources rely heavily on the information provided by chemical analysis. Manufacturing industries use analytical data to monitor the quality of raw materials, intermediates and finished products. Progress and research in many areas is dependent on establishing the chemical composition of man-made or natural materials, and the monitoring of toxic substances in the environment is of ever increasing importance. Studies of biological and other complex systems are supported by the collection of large amounts of analytical data.

④ Scope and Applications of Analytical Chemistry

Analytical data are required in a wide range of disciplines and situations that include not just chemistry and most other sciences, from biology to zoology, but the arts, such as painting and sculpture, and archaeology. Space exploration and clinical diagnosis are two quite disparate areas in which analytical data is vital. Important areas of application include the following.

1. Analytical chemistry has applications including: in forensics, bioanalysis, clinical analysis, environmental analysis, and materials analysis. Analytical chemistry research is largely driven by performance (sensitivity, detection limit, selectivity, robustness, dynamic range, linear range, accuracy, precision, and speed), and cost (purchase, operation, training, time, and space).

2. Among the main branches of contemporary analytical atomic spectrometry, the most widespread and universal are optical and mass spectrometry.
3. **Quality control (QC).** In many manufacturing industries, the chemical composition of raw materials, intermediates and finished products needs to be monitored to ensure satisfactory quality and consistency. Virtually all consumer products from automobiles to clothing, pharmaceuticals and foodstuffs, electrical goods, sports equipment and horticultural products rely, in part, on chemical analysis. The food, pharmaceutical and water industries in particular have stringent requirements backed by legislation for major components and permitted levels of impurities or contaminants. The electronics industry needs analyses at ultra-trace levels (parts per billion) in relation to the manufacture of semi-conductor materials. Automated, computer-controlled procedures for process-stream analysis are employed in some industries.
4. **Monitoring and control of pollutants.** The presence of toxic heavy metals (e.g., lead, cadmium and mercury), organic chemicals (e.g., polychlorinated biphenyls and detergents) and vehicle exhaust gases (oxides of carbon, nitrogen and sulfur, and hydrocarbons) in the environment are health hazards that need to be monitored by sensitive and accurate methods of analysis, and remedial action taken. Major sources of pollution are gaseous, solid and liquid wastes that are discharged or dumped from industrial sites, and vehicle exhaust gases.
5. **Clinical and biological studies.** The levels of important nutrients, including trace metals (e.g., sodium, potassium, calcium and zinc), naturally produced chemicals, such as cholesterol, sugars and urea, and administered drugs in the body fluids of patients undergoing hospital treatment require monitoring. Speed of analysis is often a crucial factor and automated procedures have been designed for such analyses.
6. **Geological assays.** The commercial value of ores and minerals is determined by the levels of particular metals, which must be accurately established. Highly accurate and reliable analytical procedures must be used for this purpose, and referee laboratories are sometimes employed where disputes arise.
7. **Fundamental and applied research.** The chemical composition and structure of materials used in or developed during research programs in numerous disciplines can be of significance. Where new drugs or materials with potential commercial value are synthesized, a complete chemical characterization may be required involving considerable analytical work. Combinatorial chemistry is an approach used in pharmaceutical research that generates very large numbers of new compounds requiring confirmation of identity and structure.
8. In the *direct elemental analysis of solid samples*, the new leaders are laser-induced breakdown and laser ablation mass spectrometry, and the related techniques with transfer of the laser ablation products into inductively coupled plasma. Advances in design of diode lasers and optical parametric oscillators promote developments in fluorescence and ionization spectrometry and also in absorption techniques where uses of optical cavities for increased effective absorption path length are expected to expand. The use of plasma- and laser-based methods is increasing.
9. An interest towards *absolute (standardless) analysis* has revived, particularly in emission spectrometry.
10. Great effort is being put in *shrinking the analysis techniques to chip size*. Although there are few examples of such systems competitive with traditional analysis techniques, potential advantages include size/portability, speed, and cost.

- (micro total analysis system (μ TAS) or lab-on-a-chip). Microscale chemistry reduces the amounts of chemicals used.
11. Many developments *improve the analysis of biological systems*. Examples of rapidly expanding fields in this area are genomics, DNA sequencing and related research in genetic fingerprinting and DNA microarray; proteomics, the analysis of protein concentrations and modifications, especially in response to various stressors, at various developmental stages, or in various parts of the body, metabolomics, which deals with metabolites; transcriptomics, including mRNA and associated fields; lipidomics - lipids and its associated fields; peptidomics - peptides and its associated fields; and metalomics, dealing with metal concentrations and especially with their binding to proteins and other molecules.
 12. Analytical chemistry has played critical roles in the *understanding of basic science* to a variety of practical applications, such as biomedical applications, environmental monitoring, quality control of industrial manufacturing, forensic science and so on.
 13. The recent developments of *computer automation* and information technologies have extended analytical chemistry into a number of new biological fields. For example, automated DNA sequencing machines were the basis to complete human genome projects leading to the birth of genomics. Protein identification and peptide sequencing by mass spectrometry opened a new field of proteomics.
 14. Analytical chemistry has been an indispensable area in the development of *nanotechnology*. Surface characterization instruments, electron microscopes and scanning probe microscopes enables scientists to visualize atomic structures with chemical characterizations.

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* 1.2 ANALYTICAL PROBLEMS AND THEIR SOLUTIONS

Many problems in analytical chemistry begin with the need to identify what is present in a sample. This is the scope of a *qualitative analysis*, examples of which include identifying the products of a chemical reaction, screening an athlete's urine for the presence of a performance-enhancing drug, or determining the spatial distribution of Pb on the surface of an airborne particulate. Much of the early work in analytical chemistry involved the development of *simple chemical tests* to identify inorganic ions and organic functional groups. The classical laboratory courses in inorganic and organic qualitative analysis, still taught at some schools, are based on this work.⁸ Currently, most qualitative analyses use methods such as infrared (IR) spectroscopy and nuclear magnetic resonance (NMR) spectroscopy. These qualitative applications are covered adequately elsewhere in the undergraduate curriculum and, so, will receive no further consideration in this text.

Perhaps the most common analytical problem is a *quantitative analysis*. Examples of typical quantitative analyses include the elemental analysis of a newly synthesized compound, measuring the concentration of glucose in blood, or determining the difference between the bulk and surface concentrations of Cr in steel. Much of the analytical work in clinical, pharmaceutical, environmental, and industrial labs involves developing new quantitative methods for trace amounts of chemical species in complex samples. Most of the examples in this text are quantitative analyses.

Another important area of analytical chemistry, which receives some attention in this text, is the development of new methods for characterizing physical and chemical properties. Determinations of chemical structure, equilibrium constants, particle size, and surface structure are examples of a *characterization analysis*.

The purpose of a qualitative, quantitative, or characterization analysis is to solve a problem associated with a particular sample. The purpose of a **fundamental analysis**, on the other hand, is to improve our understanding of the theory behind an analytical method. Extending and improving the theory on which an analytical method is based, studying an analytical method's limitations, and designing and modifying existing analytical method are examples of fundamental studies in analytical chemistry.

Analytical chemists work to improve the ability of all chemists to make meaningful measurements. Chemists working in the other traditional areas of chemistry, as well as in interdisciplinary fields such as medicinal chemistry, clinical chemistry, and environmental chemistry, need better tools for analyzing materials. The need to work with smaller samples, with more complex materials, with processes occurring on shorter time scales, and with species present at lower concentrations challenges analytical chemists to improve existing analytical methods and to develop new ones.

Typical problems on which analytical chemists work include qualitative analyses (What is present?), quantitative analyses (How much is present?), characterization analyses (What are the sample's chemical and physical properties?), and fundamental analyses (How does this method work and how can it be improved?).

Analytical Perspective

The stages or steps in an overall analytical procedure can be summarized as follows:

Definition of the problem: Analytical information and level of accuracy required. Costs, timing, availability of laboratory instruments and facilities.

Choice of technique and method: Selection of the best technique for the required analysis, such as chromatography, infrared spectrometry, titrimetry, thermogravimetry. Selection of the method (i.e. the detailed stepwise instructions using the selected technique).

Sampling: Selection of a small sample of the material to be analyzed. Where this is heterogeneous, special procedures need to be used to ensure that a genuinely representative sample is obtained.

Sample pre-treatment or conditioning: Conversion of the sample into a form suitable for detecting or measuring the level of the analyte(s) by the selected technique and method. This may involve dissolving it, converting the analyte(s) into a specific chemical form or separating the analyte(s) from other components of the sample (the sample matrix) that could interfere with detection or quantitative measurements.

Qualitative analysis: Tests on the sample under specified and controlled conditions. Tests on reference materials for comparison. Interpretation of the tests.

Quantitative analysis: Preparation of standards containing known amounts of the analyte(s) or of pure reagents to be reacted with the analyte(s). Calibration of instruments to determine the responses to the standards under controlled conditions. Measurement of the instrumental response for each sample under the same conditions as for the standards. All measurements may be replicated to improve the reliability of the data, but this has cost and time implications. Calculation of results and statistical evaluation.

Preparation of report or certificate of analysis: This should include a summary of the analytical procedure, the results and their statistical assessment, and details of any problems encountered at any stage during the analysis.

Review of the original problem: The results need to be discussed with regard to their significance and their relevance in solving the original problem. Sometimes repeat analyses or new analyses may be undertaken.

1.3 THE NATURE OF ANALYTICAL METHODS

week 2 Depending on the nature of analytical method it can be classified into qualitative or quantitative.

★ **Qualitative analysis** is that measured property indicates presence of analyte in matrix
Quantitative analysis is that magnitude of measured property is proportional to concentration of analyte in matrix

Species of interest: All constituents including analyte and matrix-analyte (concomitants) often need pretreatment - chemical extraction, distillation, separation, precipitation.

(A) Classical:

Qualitative - identification by color, indicators, boiling points, odors
 Quantitative - mass or volume (e.g. gravimetric, volumetric)

(B) Instrumental:

Qualitative - chromatography, electrophoresis and identification by measuring physical property (e.g. spectroscopy, electrode potential)

Quantitative - measuring property and determining relationship to concentration (e.g. spectrophotometry, mass spectrometry) often, same instrumental method used for qualitative and quantitative analysis.

Table 1.1: A general classification of important analytical methods

Group	Property measured
Gravimetric	Weight of pure analyte or of a stoichiometric compound containing it
Volumetric/Titration	Volume of standard reagent solution reacting with the analyte
Spectrometric	Intensity of electromagnetic radiation emitted or absorbed by the analyte
Electrochemical	Electrical properties of analyte solutions
Chromatographic	Physico-chemical properties of individual analytes after separation

Table 1.2: Types of instrumental methods

Property	Example Method
Radiation emission	Emission spectroscopy - fluorescence, phosphorescence, luminescence
Radiation absorption	Absorption spectroscopy - spectrophotometry, photometry, nuclear magnetic resonance, electron spin resonance
Radiation scattering	Turbidity, Raman
Radiation refraction	Refractometry, interferometry
Radiation diffraction	X-ray, electron
Radiation rotation	Polarimetry, circular dichroism
Electrical potential	Potentiometry
Electrical charge	Coulometry
Electrical current	Voltammetry - amperometry, polarography

Electrical resistance	Conductometry
Mass	Gravimetry
Mass-to-charge ratio	Mass spectrometry
Thermal	Thermal gravimetry, calorimetry
Rate of reaction	Stopped flow, flow injection analysis
Radioactivity	Activation, isotope dilution

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1.4. TRENDS IN ANALYTICAL METHODS

In the 18th and 19th centuries, there were very few professional analysts; Swedish chemist T. Bergman (1735–1784) was, probably, the first one and later, in the 19th century, Germans H. Rose, K.R. Fresenius, K. Mohr and some others should be considered as professional analytical chemists. However, many chemists of wider profile also developed early analytical chemistry, e.g., A.L. Lavoisier or Yu. Liebig. In the 20th century, separate analytical laboratories appeared and the number of professional analysts was quickly increased. At present, we have at least four groups of analysts: researchers and educators in universities; researchers and practical analysts in state services like EPA or FDA in the USA; workers in analytical laboratories in industry; agriculture and so on; developers of analytical instruments and other tools for analysis. We can think that, in the future, researchers, teachers and developers of instruments will be necessary in large numbers; as to practical analysts dealing with (routine) analyses, their numbers will decrease due to automation, wider use of continuously working sensors, and other factors.

Following trends in analytical chemistry are considered, mostly general ones: movement towards out-of-laboratory, speciation, non-destructive analysis, automatization and miniaturization, pattern recognition, moving towards more active application of achievements of physics or biochemistry. Directions of the changes in analytical techniques, e.g. their hybridization, are also discussed. There are changes even in analytical community and approaches to education.

1. Chemical analysis is more and more becoming out-of-laboratory one (moves "to the field")

Needs for such analysis are very strong. If analyses can be carried out in the field, at home, aboard a ship etc., the laboratory is becoming unnecessary. There are analyses which cannot be, in principle, done in the laboratory (methane in coal mines, chemical weapon in the field and many other cases). Many such analyses are carried out on a wide scale, e.g. determination of glucose in blood at home using portable glucometers. There are already many tools for out-of-laboratory analysis: mobile laboratory (palliative tool), portable instruments, chemical sensors, test kits, spectroscopic devices for remote analysis of atmosphere and other.

2. From the bulk analysis to the speciation analysis

That is to detection and determination of chemical and physical species of analytes. This trend is obvious in elemental analysis: we determine different oxidation states of As, Cr, Fe, Mo or different chemical forms of these and other elements, especially Hg, Sn and Se. Speciation analysis is of importance for environmental analysis, geochemistry, materials science and many other fields. Well known is, e.g., the different toxicity of various Cr, Hg or As species. Similar analytical tasks will be topical for many other elements. (By the way, "speciation analysis" is also significant for some organic compounds, for instance for substances having optical isomers of different pharmaceutical activity.)

Methodology of speciation elemental analysis is usually based on a combination of separation (chromatography) and determination (spectrometry) steps. This approach can not be considered as an ideal one; for example, a shift of equilibrium during the separation of species is possible; in this case, the pattern of species distribution will be wrong. Direct detection and determination of species is, of course, more effective; however, there are not so many techniques for the purpose. Stripping voltammetry, partially ESR and Mössbauer spectrometry are used in elemental speciation analysis (and NMR for organic compounds).

3. Trend to the non-destructive analysis of solid samples

This direction is not novel one, it was always significant in industry, archeology, forensic science, art and other areas, but the practical realization was not easy (however Archimedes carried out such an analysis in the 3th century BC). The number of techniques available for non-destructive analysis is now increasing however not fast. X-Ray methods are most important, especially XRF; in some cases, neutron activation analysis, ESR or Mössbauer spectrometry can be used.

4. Automatization of analysis

Automatization is mostly needed for the widespread analysis of untypical samples and, vice versa, not convenient for the analysis of small number of very different samples. Approaches to automatization are various for the solution of industrial and laboratory problems. In metallurgical plants, there are systems including tools for fast sampling, mail for samples transportation to a laboratory, automatic analysis using, e.g., XRF and atomic emission spectrometers, facilities for sending results to workers and tools for storing and treatment of analysis results. Automatic analytical systems in chemical and petroleum industry are usually based on gas chromatography. Automatic systems for laboratory analysis often include continuous flow (agriculture, pharma) or flow injection analysis. Laboratory robots are also used though not so often.

Everybody knows about the automation of analytical instruments (auto samplers, databases, data treatment, and control of instruments using microprocessors and so on).

5. Miniaturization of analytical tools and the analysis as a whole

Miniaturization makes it possible to decrease demands in working space, reagents, water, power etc. This trend is absolutely clear in respect to analytical devices whose size is decreasing similarly to computers. In 1960s or 1970s, for example, IR spectrometers were heavy and bulky (and sanded on the floor), now they are bench top, their dimensions decreased but the potential is growing. Mass spectrometers show the similar picture. For the last 15-20 years, a lot of mini-analyzers and more universal mini-devices, which can be considered as hand-held ones, have been developed, especially for gas analysis (not so many for water analysis). Mini-photometers, electrochemical devices, portable instruments for detection of explosives and narcotics (based mostly on ion mobility spectrometry) and even mass spectrometers can be mentioned. Pistol-form and shape instruments based on XRF and AES are widely used. Since the beginning of 1990s, microfluidic devices on microchips were developed, mostly for capillary electrophoresis, but not exclusively. The main field of their current applications is biomedical analysis, including the one based on polymerase chain reaction.

6. Change of roots.

Modern analytical chemistry borrows principles, approaches, technical decisions, tools mostly not from chemistry but from physics, biochemistry or engineering. The roots of many widely used spectrometric analytical techniques or gas sensors are located, for example, in physical science; biosensors were born on the background of biochemistry and molecular biology and so on. It is obviously that this trend is of great importance.

7. Pattern recognition

Pattern recognition is sometimes carried out instead of regular component analysis; this approach changes the paradigm of chemical analysis. Many identificational and classificational tasks can be solved without detection or determination of sample components; wine testers or perfume specialists (and, of course, dogs) identifies their object without any conventional chemical analysis. At present, such identification can be rendered more objective using sophisticated instrumentation, e.g. electronic nose or electronic tongue and chemometrical approaches.

8. Use of multistep analysis

The number of analyses is growing in many areas of application due to the wellness to improve the quality of manufactured products and service or improve the life quality. The number of components to be determined is also increased and novel objects of analysis and control are appeared. In other words, scales of analytical investigation and analytical control are permanently growing. Analysis of environmental samples can serve as an example. How analytical chemistry can answer this growing? The extensive way, that is, increasing the number of analysts, number of instruments and laboratories, is not perspective even with automatization and other modern improvements. We need principally other approaches. Using multistep analysis and control is one of the palliative ways. The main peculiarity of this is screening samples on the first (or on several first) steps. The number of samples which go to the next step can be significantly diminished.

9. Hybridization of techniques and instruments

This process is developing in several directions:

1. Combination of sample treatment and determination (flow injection analysis and its analogs);
2. Combination of separation and determination (analytical chromatography, capillary electrophoresis);
3. Combination of different determination techniques (complicated devices for surface analysis).

10. Changes of hierarchy of techniques and samples to be analyzed

Relative importance of different analytical techniques is permanently changing, and their hierarchy will not be stable in the future. The competition of methods existed always and, of course, is reasonable. Top places in scientific publications are occupied by different techniques; the changes took place, at least, every ten years. In the 1950s, the titles of publications contained mostly the words titration, precipitation, colorimeters, photometry, polarography, in the 1960s – photometry, solvent extraction, spectrochemical analysis, polarography, in the 1970s – gas chromatography and atomic absorption spectrometry, in the 1990s – liquid chromatography, solid phase extraction, in 2010 – mass spectrometry. However, quite different is the hierarchy of the number of real analyses performed by use of various analytical techniques or the numbers of instruments which have been sold. For example, AAS is practically used everywhere but the number of publications on this technique is now small.

11. Growing difficulties in education

When chemical methods of analysis predominated (19th and the first half of 20th centuries), education in analytical chemistry was a natural part of education in chemistry. At present, when analyses are mostly carried out by using physical, and partially biochemical, methods, it is necessary to find the compromise, the balance between chemistry and modern analytical methods which are often based not on chemistry. In the future, this situation will be, probably, more sharp.

12. Changes in the community of analysts

At present, there are, at least, three groups of analysts: 1) scientists and educators, 2) analysts of practical laboratories and 3) developers of analytical instruments and other tools of analysis. It is possible to predict that scientists, educators and developers will exist in a great number; the number of analysts who do routine analyses will be decreased due to automatization of analyses, application of continuously working sensors, etc.

★ 1.5. DIFFERENT UNITS OF CONCENTRATION AND THEIR CONVERSION

It will be important to understand a few terms dealing with solutions, so let's

define them:

Solution - a mixture consisting of a solute and a solvent.

Solute - component of a solution present in the lesser amount.

Solvent - component of a solution present in the greater amount.

Concentration - amount of a solute present in a solution per standard amount of solvent. There are numerous ways of expressing concentrations. Some ways to express concentration are as followed:

Weight/Weight Percent (w/w%): This unit of concentration is often used for concentrated solutions, typically acids and bases. If you were to look on a bottle of a concentrated acid or base solution the concentration is often expressed as a weigh/weight percent. A weight/weight percent is defined as:

$$w/w \% = \frac{\text{grams of solute}}{\text{grams of solution}} \times 100$$

Note: Mass percent is also another term used for weight/weight percent.

Molarity (M): This unit of concentration relates the moles of solute per liter of solution.

$$\text{Molarity} = \frac{\text{moles of solute}}{\text{L solution}}$$

Molarity is the most common concentration unit involved in calculations dealing with volumetric stoichiometry.

Molality (m): This unit of concentration relates the moles of solute per kilogram of solvent.

$$\text{Molality} = \frac{\text{moles of solute}}{\text{kg solvent}}$$

Molality is often used as the concentration unit involved in calculations dealing with colligative properties, such as freezing point depression, boiling point elevation and osmotic pressure.

Parts per million (ppm): This unit of concentration may be expressed in a number of ways. It is often used to express the concentration of very dilute solutions. The "technical" definition of parts per million is:

$$\text{ppm} = \frac{\text{grams of solute}}{\text{grams of solution}} \times 10^6$$

Since the amount of solute relative to the amount of solvent is typically very small, the density of the solution is to a first approximation the same as the density of the solvent. For this reason, parts per million may also be expressed in the following two ways:

$$\text{ppm} = \frac{\text{mg solute}}{\text{L solution}}$$

$$\text{ppm} = \frac{\text{mg solute}}{\text{kg solution}}$$

Parts per billion (ppb): This concentration unit is also used for very dilute solutions. The "technical" definition is as follows:

$$\text{ppb} = \frac{\text{grams solute}}{\text{grams of solution}} \times 10^9$$

Owing to the dilute nature of the solution, once again, the density of the solution will be about the same as the density of the solvent. Thus, we may also express parts per billion as:

$$\text{ppb} = \frac{\mu\text{g solute}}{\text{L solution}}$$

$$\text{ppb} = \frac{\mu\text{g solute}}{\text{kg solution}}$$

Mole fraction (X) and mole percent (%X): A fraction is defined as a part over a whole. Multiplying this fraction by 100 would give the percent. Thus, a mole fraction involves knowing the moles of solute or component of interest over the total moles of all components in the solution mixture:

$$X_{\text{solute}} = \frac{\text{mol solute}}{\text{total moles of all components}}$$

$$X_{\text{solute}} \% = \frac{\text{mol solute}}{\text{total moles of all components}} \times 100$$

One may find it necessary to be able to convert from one concentration unit to another. The key to solving this type of problem is to realize that you may make an assumption to get started. One may need to know the density of the solution, which would be given in the problem. Then, by using dimensional analysis, you try to get to the units of the concentration unit you are seeking to find. To get started, assume the quantity of solution found in the denominator unit of the concentration unit you are trying to convert. For example, if you are trying to convert weight/weight percent to molarity, assume 100

grams of solution. If one are trying to convert molarity to weight/weight percent, assume 1 liter of solution.

Let's look at a typical example. Suppose you are given a concentrated solution of HCl which is known to be 37.0% HCl and has a solution density of 1.19 g/mL. What is the molarity, molality and mole fraction of HCl?

Begin with the assumption of 100 g of solution. With this assumption, you now know a few other facts. In 100 g of solution, 37.0 g is due to HCl (grams of solute) and 63.0 g is due to water (grams of solvent).

To find molarity, we need to determine the moles of HCl (solute) per liter of solution. First, convert the known amount of HCl (37.0 g) to moles:

$$\text{mol HCl} = 37.0 \text{ g HCl} \times \frac{1 \text{ mol HCl}}{36.5 \text{ g HCl}} = 1.01 \text{ mol HCl}$$

Next, convert the known mass of solution, 100 g solution, to liters of solution, using the density of the solution:

$$\text{L soln} = 100 \text{ g solution} \times \frac{1 \text{ mL solution}}{1.19 \text{ g solution}} \times \frac{1 \text{ L solution}}{1000 \text{ mL solution}} = 0.0840 \text{ L solution}$$

Since the moles of solute (HCl) and volume of solution in liters is now know, calculate the molarity (M) as the moles of solute per liter of solution:

$$M = \frac{1.01 \text{ mol HCl}}{0.0849 \text{ L solution}} = 12.0 \text{ mol HCl/L solution} = 12.0 \text{ M HCl}$$

From the information above, let's find the molality of the HCl solution. The moles of solute are already known (1.01 mol HCl). We need to find the kilograms of solvent:

$$63.0 \text{ g H}_2\text{O} \times \frac{1 \text{ kg H}_2\text{O}}{1000 \text{ g H}_2\text{O}} = 0.0630 \text{ kg H}_2\text{O}$$

Since molality (m) is defined as the moles of solute per kilogram solvent, it becomes easy to find the molality:

$$m = \frac{1.01 \text{ mol HCl}}{0.0630 \text{ kg H}_2\text{O}} = 16.0 \text{ mol HCl/kg H}_2\text{O} = 16.0 \text{ m H}_2\text{O}$$

Finally, let's tackle the mole fraction of HCl. The moles of HCl are known to be 1.01 moles. We need to find the moles of H₂O:

$$\text{mol H}_2\text{O} = 63.0 \text{ g H}_2\text{O} \times \frac{1 \text{ mol H}_2\text{O}}{18.0 \text{ g H}_2\text{O}} = 3.50 \text{ mol H}_2\text{O}$$

Since the moles of solute (HCl) and moles of solvent (H₂O) are known, the mole fraction of HCl may be calculated:

$$X_{\text{HCl}} = \frac{1.01 \text{ mol HCl}}{1.01 \text{ mol HCl} + 3.50 \text{ mol H}_2\text{O}} = 0.244$$

1.6. ACCURACY AND PRECISION week 7

The numerical value obtained in every scientific measurement contains certain degree of uncertainty. In very accurate work the uncertainty may be very small, but it is still present. This uncertainty is called the error of the measurement. Accuracy relates to the closeness of approach of a single measurement, or of the average of a series of measurement, to the true value. Accuracy may also be defined as the concordance between the measured value and the true or most probable value. The true value cannot be measured exactly, however, in many cases it can be estimated very closely. By using calibrated equipment, performing the work very carefully, executing a very large number of measurements, and then applying statistics to the results, reasonable approximations of the true value can be obtained. Mostly, the true value is estimated from the results of different analysis in different laboratories.

Precision describes the closeness of approach of replicate results to a common value. Repeated measurements of the same quantity will usually not be identical, but will scatter around some common value. Precision describes the reproducibility or scatter of a series of measurements of result. Precision may also be defined as the concordance of a series of measurements of the same quantity.

Both accuracy and precision have specific meanings and implications, so they should not be used interchangeably or carelessly. To illustrate exactly what is meant by the two terms, accuracy and precision, let us examine two sets of data for the weight of a crucible. The data in set A were obtained by five different students each weighing the crucible on his own analytical balance, each using, therefore, a different set of weights. The second set B was obtained by weighing a sixth student weigh the crucible five times on his analytical balance, thus these data was obtained on the same equipment. The crucible is then weighted by a skilled analyst using ASTM-II-S-I weights, and applying all appropriate corrections to the weight obtained. He finds a value of 9.2474 g as the average of ten replicates, with no statistically significant trends or deviations in the individual weights or in the means. Thus, it is fairly safe to assume that this value is sufficiently close to the true value to be considered the correct weight of the crucible.

	Set A	Set B
Weight of the crucible	9.2463	9.2483
	9.2480	9.2481
	9.2477	9.2484
	9.2489	9.2480
	9.2455	9.2483
Average, weight, g	9.2473	9.2482

Let us now examine the data in sets A and B. The weights obtained in set B by the one operator using the same equipment are less scattered, much more closer to each other than those found in set A. The average of A weights obtained by the five students (Set A) is closer to the true value than is the average of weights in set B. These data reveal an interrelation between accuracy and precision that is very important. Precise data need not necessarily be accurate. A constant and reproducible error can be contained in each piece of data. The data in Set A are misleading, for it is highly unlikely that poor precision will accompany good accuracy. Precision always accompanies accuracy, but a high degree of precision does not imply accuracy.

Method for the expressing accuracy. The usual expression of accuracy is merely the difference between the piece of data (or the average of a set of data and the true value).

This is known as the *absolute error* or the *mean error*. Generally, the absolute error by itself is of low information and of little use unless one knows the magnitude of the quantity being measured or calculated. Thus saying that there is 3.0 mg low error in weighing sample, while having some value, takes on a much greater significance if we know what the weight of the object is. It makes a difference whether the weight is 30 g or 30 mg. The error is usually negligible in 30 g, but is very significant in 30 mg.

Thus the accuracy of a measurement is usually expressed in relative terms, as a fraction of the true value. It is expressed in % or in parts per thousands (ppt).

$$\text{Relative accuracy/Relative error} = \text{Absolute error/True value} \times 1000$$

$$\text{Relative accuracy in the data A} = -0.0001/9.2474 \times 1000 = -0.01 \text{ ppt}$$

$$\text{Relative accuracy in the data B} = +0.0008/9.2474 \times 1000 = +2.086 \text{ ppt}$$

It should be pointed out that as a general rule, relative accuracy and relative precision expressions are rounded off to the first decimal place. Note that smaller the numerical value of the relative error (disregarding the sign) the more accurate the data.

Thus the data in set B are less accurate than in set A.

Method of the expression of precision. There are many different ways to express the precision of a set of a data. Of these, only a few have any statistical basis, causing these expressions now to be recommended. Today these are the most frequently encountered terms. A couple of the terms below are listed for the historical purpose.

The Rang (W). This is merely the numerical difference between the highest and the lowest value of a set of results. For the data in Set A, $W = 9.2489 - 9.2455 = 0.0034 \text{ g}$, use of the range is not particularly informative since it tells nothing about the distribution of the data. The data could be distributed uniformly between the two limits or they all could be clustered around one value, with just a single datum way off. The range is most frequently used for very small set of data, i.e., two or three items. The range becomes less informative as the number of pieces of data in a set increases. The range is more informative about the precision of a set of data than the average deviation when there are fewer than eight pieces in the set.

The other methods for expressing precision are based on the absolute difference between each particular piece of data and the average of the set. This difference is known as the *deviation* of a particular piece of data. A deviation has no sign, being obtained by subtracting the datum from the average of the set. Thus for any given measurement X_i , in a finite set of data whose average arithmetic mean is m , the deviation is $= [X_i - m]$.

The mean, of course is $\Sigma_i X_i/n$,

Where n is the number of pieces of data in the set. The deviation in the sets A and B are given below:

Deviation for set A	Deviation for set B
0.0010	0.0001
0.0007	0.0001
0.0004	0.0002
0.0016	0.0002
0.0018	0.0001

1.7. RECENT DEVELOPMENTS IN THE SAMPLING TECHNIQUES

Analysis of the properties of a any material depends on the successful completion of a number of different steps: planning (identifying the most appropriate analytical

1.8. STATISTICAL ANALYSIS

Statistics is the science of making effective use of numerical data relating to groups of individuals or experiments. It deals with all aspects of this, including not only the collection, analysis and interpretation of such data, but also the planning of the collection of data, in terms of the design of surveys and experiments.

Scope

Statistics is considered by some to be a mathematical science pertaining to the collection, analysis, interpretation or explanation, and presentation of data while others consider it to be a branch of mathematics concerned with collecting and interpreting data. Because of its empirical roots and its focus on applications, statistics is usually considered to be a distinct mathematical science rather than a branch of mathematics. Statisticians improve the quality of data with the design of experiments and survey sampling. Statistics also provides tools for prediction and forecasting using data and statistical models. Statistics is applicable to a wide variety of academic disciplines, including natural and social sciences, government, and business.

Types of statistical analysis

- **Descriptive statistics** summarize the population data by describing what was observed in the sample numerically or graphically. Numerical descriptors include mean and standard deviation for continuous data types (like heights or weights), while frequency and percentage are more useful in terms of describing categorical data (like race).
- **Inferential statistics** uses patterns in the sample data to draw inferences about the population represented, accounting for randomness. These inferences may take the form of: answering yes/no questions about the data (hypothesis testing), estimating numerical characteristics of the data (estimation), describing associations within the data (correlation), modeling relationships within the data (regression), extrapolation, interpolation, or other modeling techniques like ANOVA, time series, and data mining.

In applying statistics to a scientific, industrial, or societal problem, it is necessary to begin with a **population** or process to be studied. Populations can be diverse topics such as "all persons living in a country" or "every atom composing a crystal". A population can also be composed of observations of a process at various times, with the data from each observation serving as a different member of the overall group. Data collected about this kind of "population" constitutes what is called a **time series**.

For practical reasons, a chosen subset of the population called a **sample** is studied — as opposed to compiling data about the entire group. Once a sample that is representative of the population is determined, data is collected for the sample members in an observational or experimental setting. This data can then be subjected to statistical analysis, serving two related purposes: description and inference.

Random and Systematic Errors week 7

All experimental uncertainty is due to either random errors or systematic errors. **Random errors** are statistical fluctuations (in either direction) in the measured data due to the precision limitations of the measurement device. Random errors usually result from the experimenter's inability to take the same measurement in exactly the same way to get exact the same number. **Systematic errors**, by contrast, are reproducible inaccuracies

that are consistently in the same direction. Systematic errors are often due to a problem which persists throughout the entire experiment. Note that systematic and random errors refer to problems associated with making measurements. **Mistakes** made in the calculations or in reading the instrument **are not considered in error analysis**. It is assumed that the experimenters are careful and competent.

Systematic errors

Systematic errors arise from a flaw in the measurement scheme which is repeated each time a measurement is made. If you do the same thing wrong each time you make the measurement, your measurement will differ systematically (that is, in the same direction each time) from the correct result. Some sources of systematic error are:

- Errors in the calibration of the measuring instruments.
- Incorrect measuring technique: For example, one might make an incorrect scale reading because of parallax error.
- Bias of the experimenter. The experimenter might consistently read an instrument incorrectly, or might let knowledge of the expected value of a result influence the measurements.

It is clear that systematic errors do not **average** to zero if you average many measurements. If a systematic error is discovered, a correction can be made to the data for this error. If you measure a voltage with a meter that **later** turns out to have a 0.2 V offset, you can correct the originally determined voltages by this amount and eliminate the error. Although random errors can be handled more or less routinely, there is no prescribed way to find systematic errors. One must simply sit down and think about all of the possible sources of error in a given measurement, and then do small experiments to see if these sources are active. The goal of a good experiment is to reduce the systematic errors to a value smaller than the random errors. For example a meter stick should have been manufactured such that the millimeter markings are positioned much more accurately than one millimeter.

Random errors

Random errors arise from the fluctuations that are most easily observed by making multiple trials of a given measurement. For example, if you were to measure the period of a pendulum many times with a stop watch, you would find that your measurements were not always the same. The main source of these fluctuations would probably be the difficulty of judging exactly when the pendulum came to a given point in its motion, and in starting and stopping the stop watch at the time that you judge. Since you would not get the same value of the period each time that you try to measure it, your result is obviously uncertain. There are several common sources of such random uncertainties in the type of experiments that you are likely to perform:

- Uncontrollable fluctuations in initial conditions in the measurements. Such fluctuations are the main reason why, no matter how skilled the player, no individual can toss a basketball from the free throw line through the hoop each and every time, guaranteed. Small variations in launch conditions or air motion cause the trajectory to vary and the ball misses the hoop.
- Limitations imposed by the precision of your measuring apparatus, and the uncertainty in interpolating between the smallest divisions. The precision simply means the smallest amount that can be measured directly. A typical meter stick is subdivided into millimeters and its precision is thus one millimeter.
- Lack of precise definition of the quantity being measured. The length of a table in the laboratory is not well defined after it has suffered years of use. You would find

different lengths if you measured at different points on the table. Another possibility is that the quantity being measured also depends on an uncontrolled variable. (The temperature of the object for example).

- Sometimes the quantity you measure is well defined but is subject to inherent random fluctuations. Such fluctuations may be of a quantum nature or arise from the fact that the values of the quantity being measured are determined by the statistical behavior of a large number of particles. Another example is AC noise causing the needle of a voltmeter to fluctuate.

No matter what the source of the uncertainty, to be labeled "random" an uncertainty must have the property that the fluctuations from some "true" value are equally likely to be positive or negative. This fact gives us a key for understanding what to do about random errors. You could make a large number of measurements, and average the result. If the uncertainties are really equally likely to be positive or negative, you would expect that the average of a large number of measurements would be very near to the correct value of the quantity measured, since positive and negative fluctuations would tend to cancel each other.

Estimating random errors

There are several ways to make a reasonable estimate of the random error in a particular measurement. The best way is to make a series of measurements of a given quantity (say, x) and calculate the mean $\bar{x}$, and the standard deviation σ_x from this data. The mean is defined as

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$$

where x_i is the result of the i^{th} measurement and N is the number of measurements. The standard deviation is given by

$$\sigma_x = \left(\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2 \right)^{1/2}$$

If a measurement (which is subject only to random fluctuations) is repeated many times, approximately 68% of the measured values will fall in the range $\bar{x} \pm \sigma_x$.

We become more certain that $\bar{x}$ is an accurate representation of the true value of the quantity x the more we repeat the measurement. A useful quantity is therefore the standard deviation of the mean $\sigma_{\bar{x}}$ defined as $\sigma_{\bar{x}} \equiv \sigma_x / \sqrt{N}$. The quantity $\sigma_{\bar{x}}$ is a good estimate of our uncertainty in $\bar{x}$. Notice that the measurement precision increases in proportion to $\sqrt{N}$ as we increase the number of measurements. Not only have you made a more accurate determination of the value, you also have a set of data that will allow you to estimate the uncertainty in your measurement.

week 10

1.9. SIGNIFICANT FIGURES AND ROUNDING OFF THE DATA

Significant figures are the reliable digits in a number or measurement which are known with certainty. Significant figures show the accuracy in measurements. We can understand the precision of a measurement if we know exactly the significant figures in the measurement. A measurement that contains more number of significant figures is more accurate than a measurement that contains less number of Significant figures. For

example: Radius of a ball is 3.3679 cm and that of the other is 3.36 cm. In this situation the first measurement is the most accurate as it has more number of significant figures.

A number used to express the result of a physical measurement or a calculation should contain sufficient digits to express the accuracy of the measurement or calculation. The general rule used in chemistry states that the least digit of a number, has certain uncertainty in it, while all of the other digits in the number contains no uncertainty. To illustrate, a sample is weighed on an analytical balance that has accuracy ± 0.1 mg. To report its weight as 1.4063 g is incorrect since the last decimal place is meaningless, because the balance is only capable of giving accurate information to the fourth decimal place. The weight should be properly written as 1.4064 g. Using the number of significant figures, the last digit, the 4 is uncertain by ± 1 and this accuracy represents the capability of the balance. It is equally incorrect to express the sample weight as 1.406 g for in weighing the sample; there is no uncertainty about the number of milligrams taken. Full advantage of the accuracy and sensitivity of a measuring instrument should be taken and the results obtained with it should reflect this accuracy.

Let us try to determine the correct number of significant figures in a number, 0.4060 g. How many significant figures does it have? Four, the zero to the left of the decimal point is merely to indicate the order of magnitude, to show that the weight is less than one gram. In no way it is a part of the number. The other two zeros are significant figures. As a general rule for numbers smaller than one, zeros preceding the first non-zero digit are not significant, but merely indicate order of magnitude of the number. Use of the exponential form for writing numbers eliminates confusion. Thus 0.004103 contains four significant figures, three zeros preceding the 4 are non significant. This can be seen more clearly when the number is written as 4.103×10^{-3} . When writing numbers larger than one, the use of the exponential form is essential for clarity. Zeros to the right of the last non zero digit of a number larger than one, can be either significant or not; there is no way for the person reading the number to decide. Consider the number written as 4,510,000. How many significant figures does it contain? Three? Four? Seven? When the number is written like this there is no way to judge. Some, perhaps all, of the zeros to the right of 1 may be merely to locate the decimal point. Some may be all, may be significant figures. In order to eliminate this confusion, the number should be written as say 4.510×10^6 , which states clearly that there are four significant figures and still locate the decimal point.

It is customary at all intermediate stages of calculations to carry along one more significant figure than is proper. The calculation is rounded off to the correct number of significant figures only at the very end. This practice causes the error in the final result to be an accurate reflection of the final errors in the components. Only the final digit will contain a minimum of uncertainty.

In performing calculations one must be constantly alert so that the numerical result does not reflect an accuracy that is greater or less than the least accurate number used in the calculation. When using computers, desk calculators or even with hand calculations, it is very easy to obtain long strings of digit. It is deceptive to believe these really reflect accuracy. In performing calculations we must be sure to eliminate all meaningless, deceptive, excess digits from the results. The operational rules used for addition and subtraction are different from those used for multiplication and division. In addition and subtraction, the number of decimal place limit is the number of significant figures whereas in multiplication and division the number of decimal place is irrelevant. The result is restricted by the relative accuracy of the numbers involved in the

calculation. The precision and accuracy of an experiment can never be increased by the computational process. Under certain conditions they may be decreased.

Rules of Significant Figures

In order to determine significant figures in a number we must follow the following rules:

- (1) All the non-zero digits are significant figures. For Example: 3.456 have four significant figures. 12.3456 have six significant figures. 0.34 has two significant figures.
- (2) Zeros between non-zero digits are significant. For Example: 2306 has four significant figures. 20,0894 has six significant figures.
- (3) Zeros locating the position of decimal in numbers of magnitude less than one are not significant. For Example: 0.2224 has only one significant figures. 0.0000034 has two significant figures.
- (4) Final zeros to the right of the decimal point are significant. For Example: 3.0000 have five significant figures. 1002.00 have six significant figures.
- (5) Zeros that locate decimal point in numbers greater than one are not significant. For Example: 30000 have only one significant figure. 120000 have two significant figures.

Addition and Subtraction

In addition and subtraction, the number of significant figures in the sum or difference can change from the number of significant figures of the components. Only the last digit of the sum or difference may contain any uncertainty. A sum or difference may contain no more decimal place than that component number having the fewest decimal place. Consider each of the following additions.

1	2	3
43.1	0.0025	10.414
4.31	4.1167	0.0037
0.431	5.9071	9.02
Sum	10.0263	19.43

In the first, all three numbers have identical significant figures, namely three, but 43.1 has the fewest decimal place and hence limit is the numbers of significant figures in the answer. Only one decimal place can be used, so that 47.8 is the correct sum. In the second sum, all three number have the same number of decimal places, even though the first one has the fewest significant figures. The result can be expressed to four decimal places. Note that in this case the sum has one more significant figures than any number. In the third sum, all three numbers have different numbers of significant figures the sum is limited to two decimal places by 9.02. The number 0.0037 cannot be included in the sum because it lies beyond the limit of the accuracy of the least accurate of the numbers. Nothing can be added beyond the second decimal place.

The same general consideration applies to subtraction.

	1	2	3
	<u>44.341</u>	<u>19.4197</u>	<u>4.96</u>
	-4.4432	-19.4153	-0.00321
Difference	<u>39.898</u>	<u>0.0044</u>	<u>4.96</u>

The first difference is limited to three decimal place by 44.341. Note that in the second subtraction, the difference has only two significant figures, whereas each of the

$$S = \sqrt{\frac{11 \times 10^{-8}}{4}}$$

$$= 2 \times 10^{-4} = 0.0002 \text{ g}$$

The smaller the value of S , the more precise the data. Note that the numerical value of the S is greater than of the average deviations for the data in both of the sets. This is due to the increased weight given to the values with larger deviations.

The variance S^2 . For certain statistical tests it is more convenient to use the square of the standard deviation, instead of the standard deviation. This term is called the variance, and it is defined as the sum of the square of the deviations divided by one less than the number of pieces of the data or

$$S^2 = \sum_i (X_i - \mu)^2 / n - 1$$

The variance of the data in set A is 1.96×10^{-6} , and the variance of the set of the data in B = 4×10^{-8} .

1.14. RELATIVE STANDARD DEVIATION (COEFFICIENT OF VARIATION)

As pointed out above, when discussing accuracy, the absolute value of the standard deviation is of limited value. Usually the value of the mean of the data set is given along with the standard deviation which does enable one to estimate the relative magnitude of the S . When the S is expressed as a fraction of the mean of the set of data, a better feeling for the proportionality of the error is obtained. The relative S is nothing more than the S divided by the mean of the set. Again it is expressed in parts per thousand in order to avoid the possible confusion by the use of % values.

$$\text{Relative Standard Deviation} = s / \bar{x} \times 1000$$

$$\text{For the set A; } 0.0014 / 9.2473 \times 1000 = 0.15 \text{ ppt.}$$

$$\text{For the data in set B } = 0.0002 / 9.2482 \times 1000 = 0.02 \text{ ppt}$$

Thus, once again it is found that the data in set B is more precise than those in the set A. The smaller the value of the relative standard deviation.

1.15. CONFIDENCE LIMITS week 9

The true or accepted mean of a set of experimental results is generally unknown except where a certified reference material is being checked or analyzed for calibration purposes. In all other cases, an estimate of the accuracy of the experimental mean, $\bar{x}$, must be made. This can be done by defining a range of values on either side of $\bar{x}$ within which the true mean, μ , is expected to lie with a defined level of probability. This range, which ideally should be as narrow as possible, is based on the standard deviation and is known as the confidence interval, CI, and the upper and lower limits of the range as confidence limits, CL. Confidence limits can be calculated using the standard deviation, σ , if it is known, or the estimated standard deviation, s , for the data. In either case, a probability level must be defined, otherwise the test is of no value. When the standard deviation is already known from past history, the confidence limits are given by the equation

$$CL(\mu) = \bar{x} \pm \frac{z\sigma}{\sqrt{N}}$$

where z is a statistical factor related to the probability level required, usually 90%, 95% or 99%. The values of z for these levels are 1.64, 1.96 and 2.58, respectively, and correspond

to the multiples of the standard deviation shown in Figure 1.2. Where an estimated standard deviation is to be used, δ is replaced by s , which must first be calculated from the current data. The confidence limits are then given by the equation.

$$CL(\mu) = \bar{x} \pm \frac{ts}{\sqrt{N}}$$

where z is replaced by an alternative statistical factor, t , also related to the probability level but in addition determined by the number of degrees of freedom for the set of data, i.e. one less than the number of results. It should be noted that (i) the confidence interval is inversely proportional to $\sqrt{N}$, and (ii) the higher the selected probability level, the greater the confidence interval becomes as both z and t increase. A probability level of 100 percent is meaningless, as the confidence limits would then have to be $\pm \infty$. The following examples demonstrate the calculation of confidence limits using each of the two formulae.

Example 1.7: The chloride content of water samples has been determined a very large number of times using a particular method, and the standard deviation found to be 7 ppm. Further analysis of a particular sample gave experimental values of 350 ppm for a single determination, for the mean of two replicates and for the mean of four replicates.

Using equation $CL(\mu) = \bar{x} \pm \frac{z\sigma}{\sqrt{N}}$, and at the 95% probability level, $z = 1.96$ and the confidence limits are:

$$1 \text{ determination } CL(\mu) = 350 \pm \frac{1.96 \times 7}{\sqrt{1}} = 350 \pm 14 \text{ ppm}$$

$$2 \text{ determination } CL(\mu) = 350 \pm \frac{1.96 \times 7}{\sqrt{2}} = 350 \pm 10 \text{ ppm}$$

$$4 \text{ determination } CL(\mu) = 350 \pm \frac{1.96 \times 7}{\sqrt{4}} = 350 \pm 7 \text{ ppm}$$

Example 1.8: The same chloride analysis as in above example, but using a new method for which the standard deviation was not known, gave the following replicate results, mean and estimated standard deviation:

Chloride/ppm	Mean	Estimated standard deviation
346	351.67 ppm	6.66 ppm
359		
350		

Using equation, $CL(\mu) = \bar{x} \pm \frac{ts}{\sqrt{N}}$ and at the 95% probability level, $t = 4.3$ for two degrees of freedom, and the confidence limits are:

$$3 \text{ determinations } CL(\mu) = 352 \pm \frac{4.3 \times 6.66}{\sqrt{3}} = 352 \pm 17 \text{ ppm}$$

The wider limits given by equation $CL(\mu) = \bar{x} \pm \frac{ts}{\sqrt{N}}$ when the standard

deviation is estimated with only three results reflects the much greater uncertainty associated with this value, which in turn affects the confidence in the degree of accuracy. To demonstrate good accuracy the confidence interval, CI, should be as small as possible and increasing the number of replicates will clearly achieve this. However, due to the $\sqrt{N}$

term in the denominator, to reduce the interval by, say, a factor of two requires an increase in the number of replicates by a factor of four as shown by Example 1.7. Unfortunately, the law of diminishing returns applies here, so if the CI is to be halved again, the number of replicates must be increased from four to sixteen. Similarly, in Example 1.8, the number of replicates would have to be increased from three to twelve to halve the CI, which would represent an unacceptable amount of time and money for most analytical laboratories.

1.16. SIGNIFICANCE TESTS

Significance tests involve a comparison between a calculated experimental factor and a tabulated factor determined by the number of values in the set(s) of experimental data and a selected probability level that the conclusion is correct. They are used for several purposes, such as:

- to check individual values in a set of data for the presence of determinate errors (bias);
- to compare the precision of two or more sets of data using their variances;
- to compare the means of two or more sets of data with one another or with known values to establish levels of accuracy.

Tests are based on a *null hypothesis* (H_0): an assumption that there is no significant difference between the values being compared. The hypothesis is accepted if the calculated experimental factor is less than the corresponding tabulated factor, otherwise it is rejected and there is said to be a significant difference between the values at the selected probability level. The conclusion should always be stated clearly and unambiguously. Probability levels of 90%, 95% and 99% are generally considered appropriate for most purposes, but it should be remembered that there are also corresponding 10%, 5% or 1% probabilities, respectively, of the opposite conclusion being valid. For example, if a test indicates that the null hypothesis is correct and that there is no significant difference between two values at the 95% probability level, it also allows the possibility that there is a significant difference at the 5% level.

Separate tabular values for some significance test factors have been compiled for what are described as one-tailed and two-tailed tests. The exact purpose of the comparison that is to be made determines which table to use.

- The one-tailed test is used EITHER to establish whether one experimental value is significantly greater than the other OR the other way around.
- The two-tailed test is used to establish whether there is a significant difference between the two values being compared, whether one is higher or lower than the other not being specified.

1.17. STUDENT *t*-TEST (COMPARISON OF TWO MEANS)

Among the most commonly used statistical significance tests applied to small data sets (population's samples) is the series of Student's tests. One of these tests is used for the comparison of two means, which is commonly applied to many cases. Typical examples are:

Example 1: Comparison of analytical results obtained with the same method on samples A and B, in order to confirm whether both samples contain the same percentage of the measured analyte or not.

Example 2: Comparison of analytical results obtained with two different methods A and B on the same sample, in order to confirm whether both methods provide similar analytical results or not.

Student's t-test for the comparison of two means

This test (as described below) assumes: (a) A normal (Gaussian) distribution for the populations of the random errors, (b) there is no significant difference between the standard deviations of both population samples. The two means and the corresponding standard deviations are calculated by using the following equations (n_A and n_B are the number of measurements in data set A and data set B, respectively):

$$\bar{x}_A = \sum_{i=1}^{n_A} x_i / n_A \quad \bar{x}_B = \sum_{i=1}^{n_B} x_i / n_B$$

$$S_A = \sqrt{\frac{\sum_{i=1}^{n_A} (\bar{x}_A - x_i)^2}{n_A - 1}} \quad S_B = \sqrt{\frac{\sum_{i=1}^{n_B} (\bar{x}_B - x_i)^2}{n_B - 1}}$$

Then, the pooled estimate of standard deviation S_{AB} is calculated:

$$S_{AB} = \sqrt{\frac{(n_A - 1)S_A^2 + (n_B - 1)S_B^2}{n_A + n_B - 2}}$$

Finally, the statistic t_{exp} (experimental t value) is calculated:

$$t_{\text{exp}} = \frac{|\bar{x}_A - \bar{x}_B|}{S_{AB} \sqrt{\frac{1}{n_A} + \frac{1}{n_B}}}$$

t_{exp} value is compared with the critical (theoretical) t_{th} value corresponding to the given degree of freedom N (in the present case $N = n_A + n_B - 2$) and the confidence level chosen. Tables of critical t values can be found in any book of statistical analysis, as well as in many quantitative analysis textbooks. If $t_{\text{exp}} > t_{\text{th}}$ then H_0 is rejected else H_0 is retained.

1.18. F-TEST

The significance tests described so far are used for comparing means, and hence for detecting systematic errors. In many cases it is also important to compare the standard deviations, i.e. the random errors of two sets of data. As with tests on means, this comparison can take two forms. Either we may wish to test whether Method A is more precise than Method B or we may wish to test whether Methods A and B differ in their precision. For example, if we wished to test whether a new analytical method is more precise than a standard method we would use a one-sided test; if we wished to test whether two standard deviations differ significantly a two-sided test would be appropriate. The F-test uses the ratio of the two sample variances, i.e. the ratio of the squares of the standard deviations, s_1^2/s_2^2 . The F-test may be used to decide whether there is sufficient reason to believe that two estimates of a standard deviation differ

Statistical analysis of results:**Hypothesis;****Null Hypothesis:** H_0 $\mu_1 = \mu_2 = \mu_3$ **Alternate Hypothesis:** H_1 $\mu_1 \neq \mu_2 \neq \mu_3$ **Confidence level:** $\alpha = 0.05$ **Statistical test applied:** One Way ANOVA**Computations:** Using SPSS17 software

Table 1.6: One way ANOVA

ANOVA					
Concentration					
	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	.000	2	0.000	1.938	0.150
Within Groups	.003	87	0.000		
Total	.003	89			

Results Interpretation:

Null hypothesis is accepted because P-value (0.15) is greater than α value. So there is no significant difference between these three methods, i.e., AAS, ICPOES and cold vapor technique. Any of these methods can be applied for determination of mercury (II).

1.21. COMPUTATION OF ANALYTICAL DATA

Although the calculations in this chapter are relatively straight forward, it can be tedious to work problems using nothing more than a calculator. These calculations can be done using Microsoft Excel or any other SPSS software.

Excel: It provides two methods for working with data: built-in functions for individual statistical parameters and a package of data analysis tools in the *Analysis ToolPak*. The *ToolPak* is not a standard part of Excel's installation. To see if you have access to the *Analysis ToolPak* on your computer, select *Tools* from the menu bar and look for the *Data Analysis...* option. If you do not see *Data Analysis...*, select *Add-ins...* from the *Tools* menu. Check the box for the *Analysis ToolPak* and click on *OK* to install them.

Descriptive Statistics

Let's use Excel to provide a statistical summary of the data. Enter the data into a spreadsheet, as shown in Fig 1.4. Select *Data Analysis...* from the *Tools* menu, which opens a window entitled "*Data Analysis*." Scroll through the window, select *Descriptive Statistics* from the available options, and click *OK*. Place the cursor in the box for the "*Input Range*" and then click and drag over the cells B1:B8. Check the box for "*Labels in the first row*." Select the radio button for "*Output range*," place the cursor in the box and click on an empty cell; this is where Excel will place the results. Check the boxes for "*Summary statistics*" and for the "*Confidence level for the mean*." Accept the default value of 95% for the confidence level. Clicking *OK* generates the information shown in Fig.1.5.

	A	B
1		Mass (g)
2		3.080
3		3.094
4		3.107
5		3.056
6		3.112
7		3.174
8		3.198

Fig. 1.4: Portion of a spreadsheet containing data.

Mass (g)	
Mean	3.11728571
Standard Error	0.01924369
Median	3.107
Mode	#N/A
Standard Deviation	0.05091403
Sample Variance	0.00259224
Kurtosis	-0.59879248
Skewness	0.72905145
Range	0.142
Minimum	3.056
Maximum	3.198
Sum	21.821
Count	7
Confidence Level (95.0%)	0.04708762

Fig. 1.5: Output from Excel's *Descriptive Statistics* command in the *Analysis ToolPak*.

Note that Excel does not adjust for significant figures. The mode is the most common result, which is not relevant here. Kurtosis is a measure of the "peakedness" of the data's distribution, and is zero for a normal distribution. Skewness is a measure of the symmetry of the data's distribution and is zero for a normal distribution. For a small sample size—such as the seven samples in this data set—skewness and kurtosis are not particularly useful.

Significance Tests

Excel's *Analysis ToolPak* includes tools to help you complete the following significance tests covered in this chapter:

- F-test of variances
- t-test of sample means assuming equal variances
- t-test of sample means assuming unequal variances
- t-test for of sample means

To complete the F-test select *Data Analysis...* from the *Tools* menu, which opens a window entitled "Data Analysis." Scroll through the window, select *F-Test Two Sample Variance* from the available options, and click OK. Place the cursor in the box for the

"Variable 1 range" and then click and drag over the cells like: B1:B8. Next, place the cursor in the box for "Variable 2 range" and then click and drag over the cells like: B1:B6 as in Fig. 1.6. Check the box for "Labels." In the box for "Alpha" enter your value for α , as in Fig. 1.6. Excel's F-test is a one-tailed test, so if you wish to complete a two-tailed test, as we do in this example, divide your α value by 2. We want our F-test to use an α of 0.05, so we enter 0.025 in the box for "Alpha." Select the radio button for "Output range," place the cursor in the box and click on an empty cell; this is where Excel will place the results. Clicking OK generates the information shown in Fig. 1.7. Because F_{exp} , which is 1.87, is less than $F(0.05, 6, 4)$, which is 9.20, we retain the null hypothesis and have no evidence of a difference between the variances at an α of 0.05.

	A	B	C
1		Set 1	Set 2
2		3.080	3.052
3		3.094	3.141
4		3.107	3.083
5		3.112	3.048
6		3.112	3.048
7		3.174	
8		3.198	

Fig.1.6: Portion of a spreadsheet containing the data.

F-Test Two - Sample for Variances		
	Data Set 1	Data Set 2
Mean	3.11728571	3.0814
Variance	0.00259224	0.0013843
Observations	7	6
df	6	4
F	1.87259849	
P(F<=f) one -tail	0.28305251	
F Critical one - tail	9.19731108	

Fig.1.7: Output from Excel's F-test in the Analysis ToolPak. Note that results are for a one-tailed F-test. The value for "P(F<=f) one-tail" is the probability of incorrectly rejecting the null hypothesis of equal variances, and the value for "F Critical one-tail" is equivalent to a one-tailed $F(0.025, 6, 4)$. For a two-tailed F-test, the probability of incorrectly rejecting the null hypothesis is $2 \times P(F<=f)$, or 0.566, and F Critical is for $F(0.05, 6, 4)$.

Note: Excel assumes that the variance for variable 1 is greater than the variance for variable 2; thus, you must assign variable 1 to the data set with the larger of the two variances in order to correctly interpret the F-test.